

Multi-Agent System for Automated Cardiac MRI Analysis and Decision Support

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INTRODUCTION: Research Overview

Title: Multi-Agent System for Automated Cardiac MRI Analysis and Decision Support

- **Domain:** Clinical AI / Healthcare Informatics
- **Focus:** Multi-agent AI for automated cardiac MRI analysis, integrating image segmentation, quantitative biomarker extraction, clinical interpretation, and explainability.
- **Evaluated on ACDC and M&Ms (525 subjects):** DSC, MAE, BERTScore

INTRODUCTION: Background - Cardiovascular Disease (CVD)

CVD: ~17.9 million deaths annually (~32% of global mortality) [WHO, 2023]

- Rising prevalence of hypertension, diabetes, and obesity increases diagnostic demand [Wang et al., 2024]
- Cardiac Magnetic Resonance Imaging (CMR) is the gold standard for non-invasive cardiac morphology and function assessment [Kawel-Boehm et al., 2025]
- Superior reproducibility for left ventricular ejection fraction (LVEF), end-diastolic volume (EDV), and end-systolic volume (ESV) vs. echocardiography [Wenzel et al., 2025]
- European Society of Cardiology (ESC), American College of Cardiology (ACC), and American Heart Association (AHA) affirm CMR for dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and ischaemic heart disease.

INTRODUCTION: Background - CMR & AI Landscape

Cardiac Magnetic Resonance Imaging (CMR) studies generate hundreds of frames requiring expert multi-sequence interpretation.

- **Manual analysis:** time-consuming, expertise-dependent, inter-observer variability up to 10% [Bello et al., 2025]
- **Deep learning:** near-expert segmentation (CineMA, nnU-Net) and scar quantification (ScarNet)
- Systems operate in isolation - no integrated clinical workflow [Bello et al., 2025]
- Agentic AI enables end-to-end reasoning that is accurate, traceable, and interpretable

INTRODUCTION: Why Multi-Agent Architecture?

Multi-agent is chosen not because it is novel, but because the clinical workflow is inherently multi-step, multi-specialist, and requires verifiable handoffs, properties a single model cannot provide.

01 Inherently Decomposable Task

Cardiac MRI analysis spans 4 distinct expert subtasks, segmentation, quantification, reasoning, and explanation, each needing different models, knowledge sources, and evaluation criteria. [Gorenshtein et al., 2025]

02 Mirrors Clinical Specialisation

In practice, these tasks are performed by different specialists (radiographer → cardiologist → reporting physician → auditor). Multi-agent maps directly to that separation of responsibility. [Chen et al., 2025]

03 Failure Isolation & Safety

A single model fails silently. Multi-agent makes each step's output observable and verifiable before passing downstream, a critical requirement for clinical decision support. [Xu et al., 2026]

04 RAG Requires Dedicated Orchestration

Reasoning from ESC/ACC/AHA clinical guidelines cannot be embedded in a segmentation model. A separate reasoning agent with retrieval access is architecturally necessary. [Kohandel Gargari & Habibi, 2025]

05 Explainability as First-Class Output

XAI is not a post-hoc add-on, it is a dedicated agent. Separation of concerns is what makes explainability principled rather than retrofitted. [Frontiers in Radiology, 2025]

A monolithic end-to-end model cannot simultaneously segment, apply clinical guidelines, and produce traceable explanations, these are cognitively distinct tasks requiring architectural separation.

RESEARCH GAPS:



Knowledge Gap

No dedicated agentic system for cardiac MRI. Healthcare agents are domain-generic, lacking imaging-specific orchestration workflows [Xu et al., 2026]¹

Opportunity: Design a dedicated cardiac MRI agentic framework integrating segmentation, quantitative analysis & guideline-grounded clinical reasoning.



Experimental Gap

BAAI Cardiac Agent⁵ & CardAlc-Agents⁶ lack validation on multi-centre or heterogeneous datasets; many systems are conceptual or insufficiently validated.

Opportunity: Validate on ACDC & M&Ms datasets using standardised metrics for diagnostic accuracy and workflow alignment.



Methodological Gap

Multimodal agents — AURA², MedChat³, MMedAgent-RL⁴ — operate on isolated tasks with no unified end-to-end pipeline for complex clinical diagnosis.

Opportunity: Integrate segmentation, RAG retrieval & clinical reasoning into a single orchestrated multi-agent pipeline.



Clinical Gap

XAI reviews in cardiovascular imaging [Haupt et al.⁷; Salih et al., 2025⁸] confirm most deployed systems produce opaque predictions misaligned with ACC/AHA & ESC CMR guidelines.

Opportunity: Embed guideline-grounded outputs & attention-based explanations for transparent, traceable clinical decision support.

¹ Xu et al. (2026). Agentic AI in healthcare survey. ² Fathi, N. et al. (2025). AURA. arXiv:2507.16940 ³ Liu, P.R. et al. (2025). MedChat. MIPR 2025, 456–462. doi:10.1109/MIPR67560.2025.00078

⁴ Xia et al. (2025). MMedAgent-RL. ⁵ Qu, T. et al. (2026). BAAI Cardiac Agent. arXiv:2604.04078 ⁶ Zhang et al. (2025). CardAlc-Agents. ⁷ Haupt et al. (2025). XAI cardiovascular imaging review.

⁸ Salih et al. (2025). XAI cardiac AI review.

PROBLEM STATEMENT:



Fragmented Task Automation

CineMA (2025): Segmentation only, no clinical reasoning extension¹

MARL-Rad (2025): Structured reports not grounded in imaging findings²

AURA & MedChat (2025): Multimodal dialogue lacking cardiac imaging pipelines^{3,4,5}

No unified end-to-end CMR workflow integrating segmentation, reporting, and clinical reasoning.



Absent Guideline Grounding

BAAI Cardiac Agent & CardAIc-Agents: Outputs not aligned with ESC/ACC/AHA/HFSA guidelines¹

Patel et al. (2025): ACR RAG system restricted to general radiology text only

No CMR AI retrieves from ESC 2022 CMR or ACC/AHA cardiac imaging guidelines.

Outputs lacking traceability to guideline criteria carry limited clinical utility.



Insufficient Explainability

Majority of cardiac AI systems are functionally opaque [Atrey et al., 2025; Barros et al., 2025]

XAI mechanisms proposed but seldom validated against clinical utility criteria

Black-box predictions carry significant patient safety risk [Rudin, 2019]

Misclassified cardiomyopathy or undetected scar can alter life-altering treatment decisions.

¹ Qu, T. et al. (2026). BAAI Cardiac Agent. arXiv:2604.04078

² Fu, Y. et al. (2025). CineMA foundation model. arXiv:2506.00679

³ Baba, K. et al. (2026). MARL-Rad. arXiv:2603.16876

⁴ Fathi, N. et al. (2025). AURA. arXiv:2507.16940

⁵ Liu, P.R. et al. (2025). MedChat. MIPR 2025, 456–462. doi:10.1109/MIPR67560.2025.00078

RESEARCH QUESTIONS:

- **RQ1:** How can a multi-agent framework be developed to integrate automated segmentation, quantitative analysis, and guideline-grounded reasoning into a unified workflow that ensures clinical consistency in Cardiac Magnetic Resonance Imaging (CMR) interpretation?
- **RQ2:** How does the proposed framework improve diagnostic accuracy and provide clinically meaningful explainable reasoning compared with established single-task AI baselines using standardized metrics (Dice, MAE, and BERTScore)?
- **RQ3:** To what extent does the framework maintain diagnostic robustness and generalizability when evaluated across heterogeneous, multi-vendor datasets (ACDC and M&Ms) from diverse clinical centers?



RESEARCH OBJECTIVES:

- **O1:** To design and develop a multi-agent system that integrates automated segmentation (CineMA), quantitative analysis, and guideline-grounded reasoning into a unified cardiac MRI analysis pipeline.
- **O2:** To integrate explainable clinical reasoning into the proposed framework and benchmark its clinical accuracy and explanation quality against single-task baselines using Dice Similarity Coefficient (DSC), Mean Absolute Error (MAE), and BERTScore metrics.
- **O3:** To validate the generalizability of the proposed system across multi-center and multi-vendor heterogeneous datasets (ACDC and M&Ms) to ensure reliable performance across diverse clinical acquisition protocols.



RESEARCH NOVELTY & SCOPE: Scope Boundaries

- **In Scope:** Cine Cardiovascular Magnetic Resonance Imaging (CMR); Left Ventricle/Right Ventricle/Myocardium (LV/RV/Myo) segmentation; Left Ventricular Ejection Fraction (LVEF), End-Diastolic Volume (EDV), End-Systolic Volume (ESV); Automated Cardiac Diagnosis Challenge (ACDC) & Multi-Centre, Multi-Vendor (M&Ms) datasets
- **In Scope:** Gradient-weighted Class Activation Mapping (Grad-CAM) + guideline citations; structured clinical report generation
- **Out of Scope:** Late Gadolinium Enhancement (LGE) / T1 mapping; Right Ventricular (RV) strain; SHapley Additive exPlanations (SHAP) / Local Interpretable Model-agnostic Explanations (LIME) on raw MRI
- **Out of Scope:** Patient-facing interfaces; proprietary hospital datasets

Research prototype for decision support - adjunct to, not replacement for, specialist judgement

RESEARCH NOVELTY & SCOPE: Novelty

- First to jointly address agentic orchestration, clinical decision support, and explainability for Cardiac Magnetic Resonance Imaging (CMR)
- Curated European Society of Cardiology (ESC), American College of Cardiology (ACC), and American Heart Association (AHA) corpus specifically for cardiac imaging reasoning.
- LangGraph pipeline mirroring multi-disciplinary clinical imaging workflow.

INTRODUCTION: Research Significance

Scientific:

- First joint agentic orchestration + clinical reasoning + explainability for Cardiac Magnetic Resonance Imaging (CMR)
- Novel architecture, replicable evaluation methodology, and benchmark results

Clinical:

- Guideline-cited outputs reduce cognitive burden and improve reporting consistency

Societal:

- Supports equitable cardiac imaging access in resource-limited settings [Leiner et al., 2026]



LITERATURE REVIEW:

Thematic Domain	Key Systems & References	Collective Limitations	Research Gaps
Cardiac MRI AI	CineMA (Fu et al., 2025) ScarNet (Tavakoli et al., 2025)	Task Isolation: Fragmented models focused on segmentation without clinical reasoning	Knowledge Methodological
Cardiac AI Agents	BAAI Cardiac Agent (Qu et al., 2026) CardAIC-Agents (Zhang et al., 2025)	Lack of Alignment: No grounding in ESC/ACC guidelines; limited multi-center validation	Experimental Clinical
Multimodal Medical Agents	AURA (Fathi et al., 2025) MedChat (Liu et al., 2025)	Domain Gap: Generic healthcare frameworks; lack specialized 4D cardiac imaging pipelines	Methodological
RAG in Clinical AI	ACR RAG System (Pambudi et al., 2025)	Modality Gap: Focus exclusively on text; no extension to image-based diagnostic reasoning	Knowledge Clinical
Explainable AI (XAI)	Haupt et al. (2025) Salih et al. (2025)	Opacity: Saliency maps exist but not linked to specific clinical guideline passages	Clinical

Critical Synthesis: No existing framework simultaneously integrates: **(1) Agentic orchestration** of the CMR pipeline; **(2) Guideline-grounded reasoning** (ESC/ACC); **(3) Traceable XAI**. The proposed system fills this gap by uniting nnU-Net, FAISS, and BioBERT into a novel unified architecture.



METHODOLOGY: Research Design

- **Paradigm:** Design Science Research (DSR) - novel artefact for defined domain problem
- **Sequential mixed-methods:** quantitative metrics
- **Phase 1 - Foundation:** literature, datasets, segmentation backbone training
- **Phase 2 - Construction:** LangGraph orchestration, XAI integration
- **Phase 3 - Evaluation:** ACDC/M&Ms benchmarking, ablation, baseline comparisons
- **Phase 4 - Validation:** cross-dataset generalization testing

METHODOLOGY: System Architecture

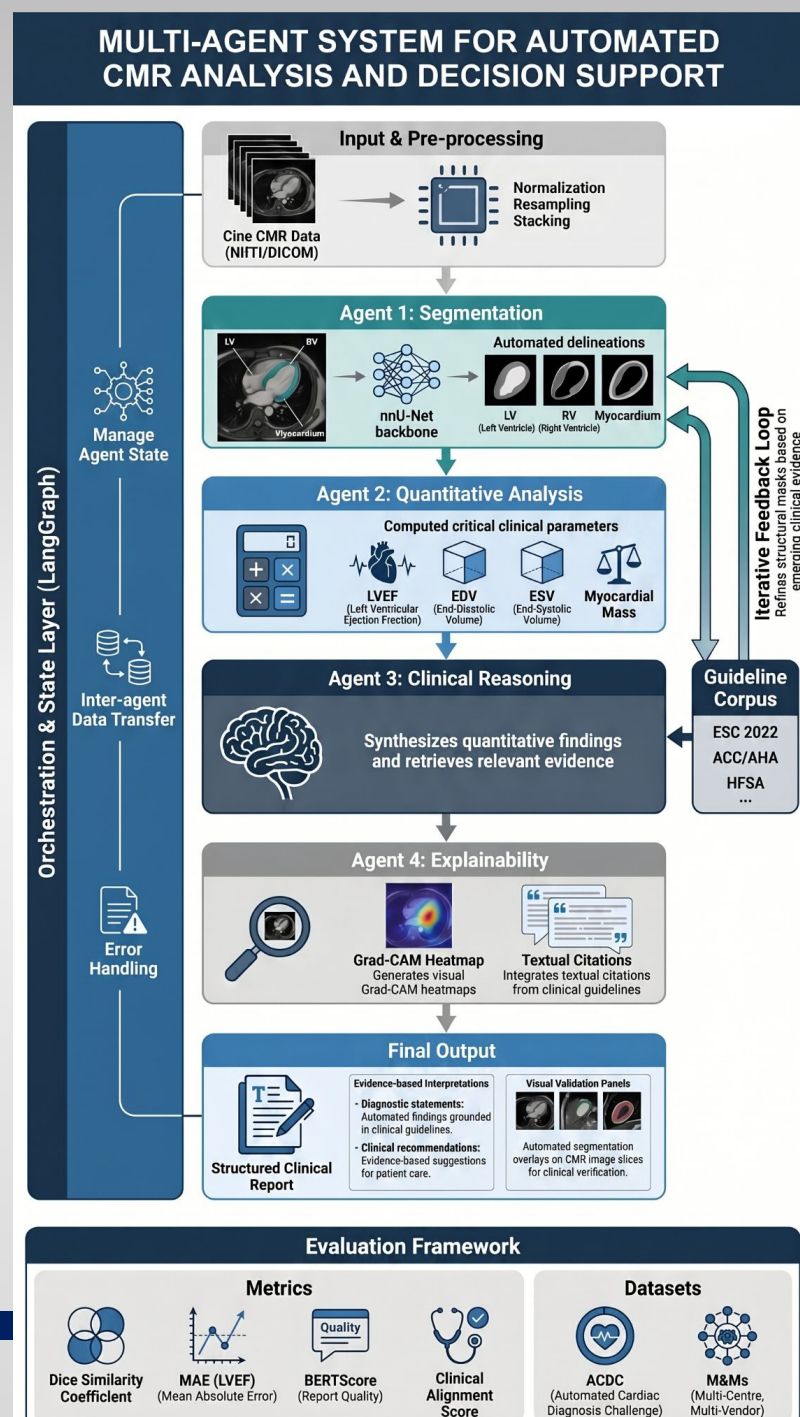
Four-agent pipeline:

Segmentation ->

Quantification ->

Clinical Reasoning ->

Explainability



METHODOLOGY: Datasets (O3)

Three publicly available datasets support training, validation, and generalisation testing.

Dataset	Year	Subjects	Classes	Modality	Purpose in This Study
ACDC	2018	150	5 pathologies	Cine CMR	Primary training & validation; segmentation + EF benchmarking
M&Ms	2021	375	4 pathologies	Cine CMR	Generalisation testing across 4 vendors and 6 clinical centres
M&Ms-2	2022	360	Multi-label	Cine CMR	Extended multi-label generalisation and stress-test evaluation

Table: Datasets used for training, validation, and generalisation testing.

ACDC — Primary Dataset

150 subjects · 5 pathology classes (DCM, HCM, MINF, ARV, Normal). Ground truth EDV, ESV, EF enable direct quantitative evaluation.

M&Ms / M&Ms-2 — Generalisation

375–360 subjects · 4 MRI vendors · 6 centres. Held out from training entirely, conservative test of domain-shift robustness (Objective O3).

METHODOLOGY: Agent 1 - Segmentation (O1) *(Phase 1 & 3)*

- Delineates LV cavity, RV cavity, and LV myocardium from cine CMR stacks
- Backbone: nnU-Net - SOTA on ACDC; CineMA as zero-shot ablation alternative
- Preprocessing: 1.5x1.5 mm resampling, z-score normalisation, ED/ES frame extraction
- Training: ACDC subjects, 5-fold CV, 1000 epochs, NVIDIA A100
- Target: Dice ≥ 0.88 (LV, RV, myocardium) on ACDC test set. [Isensee et al., 2021]

METHODOLOGY: Agent 2 - Quantitative Analysis (O1) *(Phase 2 & 3)*

- **EDV/ESV:** voxel summation across short-axis slices at ED/ES frames
- $LVEF (\%) = (EDV - ESV) / EDV \times 100$
- **LV mass:** myocardium volume x 1.05 g/mL [Kawel-Boehm et al., 2025]
- **Output:** structured JSON with ESC heart failure classification flags
- **Target:** LVEF MAE $\leq 5\%$ against expert ground truth measurements.

METHODOLOGY: Agent 3 - Clinical Reasoning (O2) *(Phase 2)*

- **Core contribution:** guideline-grounded diagnostic reasoning.
- **Corpus:** ESC Cardiac Magnetic Resonance Imaging (CMR), ACC/AHA HF, AHA/ACC Chest Pain, HFSA
- BioBERT embeddings (768-dim) + FAISS index; 200-400 token chunks, 50-token overlap
- Top-k=5 retrieval -> GPT-4o prompt -> cited rationale, differential diagnoses, recommended clinical actions
- MedLLaMA-3 ablation for open-source reproducibility

METHODOLOGY: Agent 4 - Explainability (O2) *(Phase 2)*

- **Visual:** Grad-CAM heatmaps on segmentation feature maps (ED frame overlays)
- **Textual:** guideline citations formatted into citation-annotated clinical report
- Each statement linked to source document, section, and page number
- **Dual mechanism:** interpretable visually (imaging) for radiologists and clinically (guideline language) for cardiologists.
- Directly addresses clinical gap - no existing CMR system provides both

METHODOLOGY: Orchestration Layer *(Phase 2)*

- LangGraph manages stateful, graph-based agent orchestration:
- Manage Agent State - coordinate status across all four agents
- Inter-agent Data Transfer - structured JSON between pipeline stages
- Error Handling - retry logic and graceful failure recovery
- Iterative Feedback Loop - guideline evidence can refine segmentation masks
- **Modular:** individual agents upgradeable without full architecture redesign

METHODOLOGY: Evaluation Metrics

- **Segmentation:** Dice (≥ 0.88), HD95 boundary accuracy [Isensee et al., 2021]
- **Quantification:** MAE for LVEF, EDV, ESV (LVEF MAE $\leq 5\%$)
- **Report:** BERTScore F1 (≥ 0.85)
- **Generalisation:** cross-dataset DSC drop (≤ 0.05)

EXPECTED OUTCOMES: Research Contributions

- Novel four-agent framework for end-to-end Cardiac Magnetic Resonance Imaging (CMR) interpretation
- Dual explainability: Grad-CAM + guideline-cited textual rationale
- Benchmark results on ACDC/M&Ms dataset
- Target Q1 publication: Computers in Biology and Medicine or AI in Medicine

Preliminary Findings & Research Output

- Journal Paper Submission

A key research output demonstrating the viability of the proposed framework

Submitted Paper

Transparent Cardiac Intelligence: A Post-Hoc Explainability Framework for Ensemble-Based Arrhythmia Classification

Journal: BMJ Medical Informatics & Decision Making

Submitted: 23 May 2026

Status: Under Review

Research Significance

- Integrates post-hoc XAI (SHAP / LIME) for clinical transparency.
- Demonstrates model performance on real-world cardiac dataset.
- Contributes to trustworthy AI adoption in healthcare informatics.

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METHODOLOGY: Work Schedule

Gantt Chart (07/03/2026 – 31/12/2027, 24 Months)																									
Timeline		Month 1-6						Month 7-12						Month 13-18						Month 19-24					
Months		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
No.	Activities																								
1	Literature Review & Research Framework Development																								
2	Dataset Collection, Preprocessing & Environment Setup																								
3	Development of Segmentation and Quantitative Analysis Modules																								
4	Development of Clinical Reasoning and Explainability Modules																								
5	Multi-Agent Integration & LangGraph Orchestration																								
6	System Evaluation, Benchmarking & Generalisation Testing																								
7	Thesis Writing, Journal Publication & Final Submission																								

METHODOLOGY: Ethical Considerations

- Public de-identified datasets (ACDC, M&Ms) - no institutional ethics approval required
- Research prototype only - not for clinical deployment or patient-facing use

REFERENCES

- Bello G.A. et al. (2025). Bridging the gap in cardiovascular magnetic resonance imaging artificial intelligence implementations: From ambitious goals to real-world progress using foundation models. *Journal of Cardiovascular Magnetic Resonance*. PMC12766593.
- Baba, K., Kishikawa, R., & Kodera, S. (2026). *Multi-Modal Multi-Agent Reinforcement Learning for Radiology Report Generation*. <https://doi.org/10.48550/arXiv.2603.16876>
- Bernard, O., Lalande, A., Zotti, C., Cervenansky, F., Yang, X., Heng, P.-A., Cetin, I., Lekadir, K., Camara, O., Ballester, M. A. G., Sanfilippo, F., Amic, A., Petitjean, C., & Jodoin, P.-M. (2018). Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved? *IEEE Transactions on Medical Imaging*, 37(11), 2514–2525.
- Chen, Y., Zhou, H., & Tan, W. (2025). Agentic systems in radiology: Design, applications, evaluation, and challenges. *Radiology: Artificial Intelligence*, 7(2), e240089.
- Fu, S., Dong, J., Ding, X., Sun, R., Yang, Y., Cui, S., & Li, Z. (2026). *AgentsEval: Clinically Faithful Evaluation of Medical Imaging Reports via Multi-Agent Reasoning*. <http://arxiv.org/abs/2601.16685>
- Gorenshtein, A., Omar, M., Glicksberg, B. S., Nadkarni, G. N., & Klang, E. (2025). AI agents in clinical medicine: A systematic review. *medRxiv*. <https://doi.org/10.1101/2025.08.22.25334232>
- Ganz, M., Scholten, E. T., & Bobowicz, M. (2025). Explainable AI in medicine: Challenges of integrating XAI into the future clinical routine. *Frontiers in Radiology*, 5, 1627169. <https://doi.org/10.3389/fradi.2025.1627169>
- Gorenshtein, A., Omar, M., Glicksberg, B. S., Nadkarni, G. N., & Klang, E. (2025). *AI Agents in Clinical Medicine: A Systematic Review*. <https://doi.org/10.1101/2025.08.22.25334232>
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research 1. In *Design Science in IS Research MIS Quarterly* (Vol. 28, Issue 1).
- Haupt, M., Maurer, M. H., & Thomas, R. P. (2025). Explainable Artificial Intelligence in Radiological Cardiovascular Imaging, A Systematic Review. In *Diagnostics* (Vol. 15, Issue 11). Multidisciplinary Digital Publishing Institute (MDPI). <https://doi.org/10.3390/diagnostics15111399>

REFERENCES (Continued)

- Isensee, F., Jaeger, P. F., Kohl, S. A. A., Petersen, J., & Maier-Hein, K. H. (2021). nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2), 203–211. <https://doi.org/10.1038/s41592-020-01008-z>
- Kohandel Gargari, O., & Habibi, G. (2025). Enhancing medical AI with retrieval-augmented generation: A mini narrative review. *SAGE Open Medicine*, 13. <https://doi.org/10.1177/20552076251337177>
- Kawel-Boehm, N., Hetzel, S. J., Ambale-Venkatesh, B., Captur, G., Chin, C. W. L., François, C. J., Jerosch-Herold, M., Luu, J. M., Raisi-Estabragh, Z., Starekova, J., Taylor, M., van Hout, M., & Bluemke, D. A. (2025). Society for Cardiovascular Magnetic Resonance reference values ("normal values") in cardiovascular magnetic resonance: 2025 update. *Journal of Cardiovascular Magnetic Resonance*. <https://doi.org/10.1016/j.jocmr.2025.101853>
- Kristijan CINCAR, & Todor IVAȘCU. (2025). *Artificial Intelligence-Driven Multi-Agent System for Enhanced Radiology Workflow and Decision Support*. <https://research.ebsco.com/linkprocessor/plink?id=da92e62a-0994-3ab6>
- Kolk, M. Z. H., Ruipérez-Campillo, S., Allaart, C. P., Wilde, A. A. M., Knops, R. E., Narayan, S. M., Tjong, F. V. Y., Raijmakers, F. D., van der Lingen, A. L. C. J., Götte, M. J. W., Selder, J. L., Alvarez-Florez, L., Išgum, I., & Bekkers, E. J. (2024). Multimodal explainable artificial intelligence identifies patients with non-ischaemic cardiomyopathy at risk of lethal ventricular arrhythmias. *Scientific Reports*, 14(1). <https://doi.org/10.1038/s41598-024-65357-x>
- Leiner T. et al. (2026). *The global roadmap for cardiovascular imaging: bridging the diagnostic divide*. European Heart Journal — Imaging Methods and Practice. <https://doi.org/10.1093/ehjimp/qyag031>
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240. <https://doi.org/10.1093/bioinformatics/btz682>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.

REFERENCES (Continued)

- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2020). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision*, 128(2), 336–359. <https://doi.org/10.1007/s11263-019-01228-7>
- Salih, A., Boscolo Galazzo, I., Gkontra, P., Lee, A. M., Lekadir, K., Raisi-Estabragh, Z., & Petersen, S. E. (2023). Explainable Artificial Intelligence and Cardiac Imaging: Toward More Interpretable Models. *Circulation: Cardiovascular Imaging*, 16(4). <https://doi.org/10.1161/CIRCIMAGING.122.014519>
- Sahoo, P., Tripathi, A., Saha, S., & Mondal, S. (2024). *FedMRL: Data Heterogeneity Aware Federated Multi-agent Deep Reinforcement Learning for Medical Imaging*. <http://arxiv.org/abs/2407.05800>
- Wang, Y. R., Yang, K., Wen, Y., Wang, P., Hu, Y., Lai, Y., Wang, Y., Zhao, K., Tang, S., Zhang, A., Zhan, H., Lu, M., Chen, X., Yang, S., Dong, Z., Wang, Y., Liu, H., Zhao, L., Huang, L., ... Zhao, S. (2024). Screening and diagnosis of cardiovascular disease using artificial intelligence-enabled cardiac magnetic resonance imaging. *Nature Medicine*, 30(5), 1471–1480. <https://doi.org/10.1038/s41591-024-02971-2>
- Wenzel, JP., Albrecht, JN., Toprak, B. et al. Head-to-head comparison of cardiac magnetic resonance imaging and transthoracic echocardiography in the general population (MATCH). *Clin Res Cardiol* (2025). <https://doi.org/10.1007/s00392-025-02660-1>
- Wind, S., Sopa, J., Truhn, D., Lotfinia, M., Nguyen, T.-T., Bressen, K., Adams, L., Rusu, M., Kostler, H., Wellein, G., Maier, A., & Arasteh, S. T. (2025). Agentic large language models improve retrieval-based radiology question answering. In *Lisa Adams* (Vol. 1, Issue 2).
- Xu, G., Li, X., Chen, Y., Duan, Y., Wu, S., Yu, H., Chiu, C.-H., Ni, J., Tang, N., Li, T. J.-J., Yuille, A., Jin, W., & Shi, Y. (2026). A comprehensive survey of AI agents in healthcare. *Journal of Biomedical Informatics*, 179, 105045. <https://doi.org/10.1016/j.jbi.2026.105045>

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